

Battle Cats Ultimate *Tutorial*

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It doesn't contain animation tutorial, click [here](#) for that.
Also, this guide is being rewritten, so expect changes to
be made.

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Notes

- Throughout the guide, both cat units and enemies will be collectively referred to as "units", unless explicitly stated.

Part 1. How to Download BCU

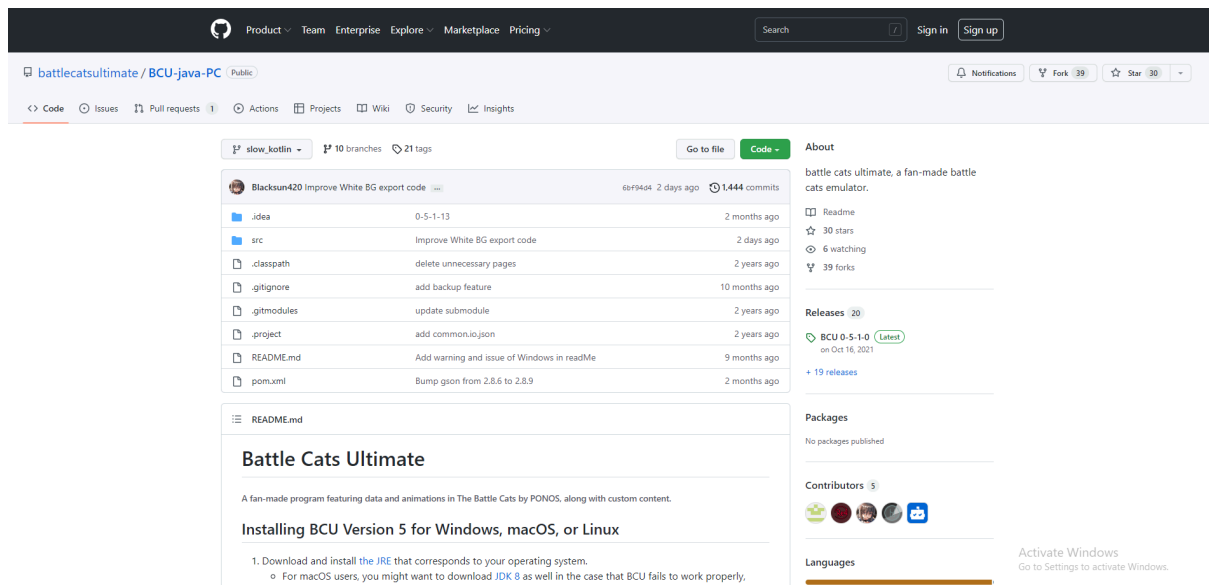
Note: You can always look in #installation-guide on the discord server.

First, download JRE from Oracle website or just click [here](#).

For Mac Users, try to download JDK too if BCU does not run properly: [Here](#)

Then, download BCU [here](#)

In the link, you will see this page



The screenshot displays the GitHub repository page for `battlecatsultimate / BCU-java-PC`. The repository is public and has 10 branches and 21 tags. The main branch is `slow_kotlin`. The repository description is "battle cats ultimate, a fan-made battle cats emulator." The repository has 30 stars, 6 watchers, and 39 forks. The latest release is `BCU 0.5-1.0` (Latest) on Oct 16, 2021. The repository has 19 releases. The README.md file is selected, showing the title "Battle Cats Ultimate" and the installation instructions for BCU Version 5 for Windows, macOS, or Linux. The instructions include downloading and installing the JRE that corresponds to the operating system, and for macOS users, downloading JDK 8 as well in case BCU fails to work properly, especially if IRVGI is enabled.

Scroll down, and select BCU-Initializer.jar

Create a new folder (name doesn't matter), and put BCU-Initializer into that folder.

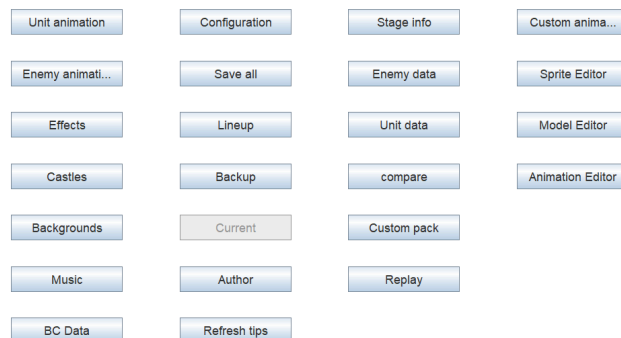
Run BCU-Initializer and it will download additional files including new BCU. It will be turned off if it downloaded all things successfully.

There will be the latest BCU file in your folder. You can use that BCU instead of BCU-Initializer.

You can delete BCU-Initializer if you want since you don't really need it, running the program automatically installs the latest version of BCU

Now you have BCU. Have fun with it! :)

memory used: 225 MB / 3554 MB, 6%
Source of enemy icon: battlecats-db.com
Author of GIF exporter: Kevin Weiner, FM Software
Welcome Hotpine!



Want to learn how to animate in BCU? Try copying some of the existing unit/enemy animations and see how they work! Or try making a simple anim and fiddling with the values.

There may be a problem that says "not enough memory" error whenever you open BCU. This is because your computer has a too small RAM size to

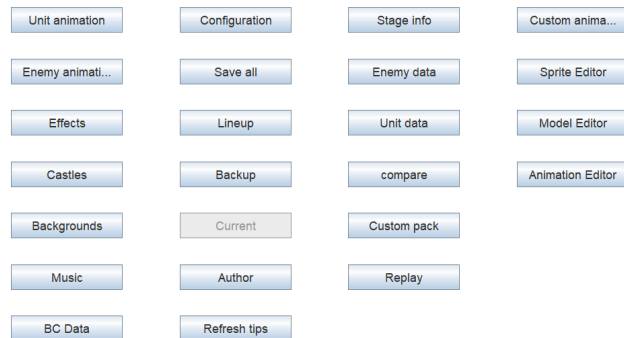
run BCU properly, but don't worry. We have a solution for this.

There is a special file called "run.bat" and you can download it [here](#). Put this file into a folder where BCU exists and run this batch file instead of BCU. You will always have to use run.bat instead of BCU from now on. Also, there must be only one BCU jar file to use run.bat properly, so if there is an update for a new BCU, remove the old jar file and leave the latest one, keeping old BCU versions and using run.bat will cause some errors.

Part 2. Getting Started

Let's use BCU now! :D

memory used: 225 MB / 3554 MB, 6%
Source of enemy icon: battlecats-db.com
Author of GIF exporter: Kevin Weiner, FM Software
Welcome Hotpine!



Want to learn how to animate in BCU? Try copying some of the existing unit/enemy animations and see how they work! Or try making a simple anim and fiddling with the values.

The first column of buttons are related to preexisting BC data

The second column is about the configuration of BCU, as well as the user interactables.

The third one is about stats and BCU packs, this tutorial will focus on this place.

The fourth one is about custom animations. If you want to make custom units, you will need this.

Now let's start talking about the basics of animation.

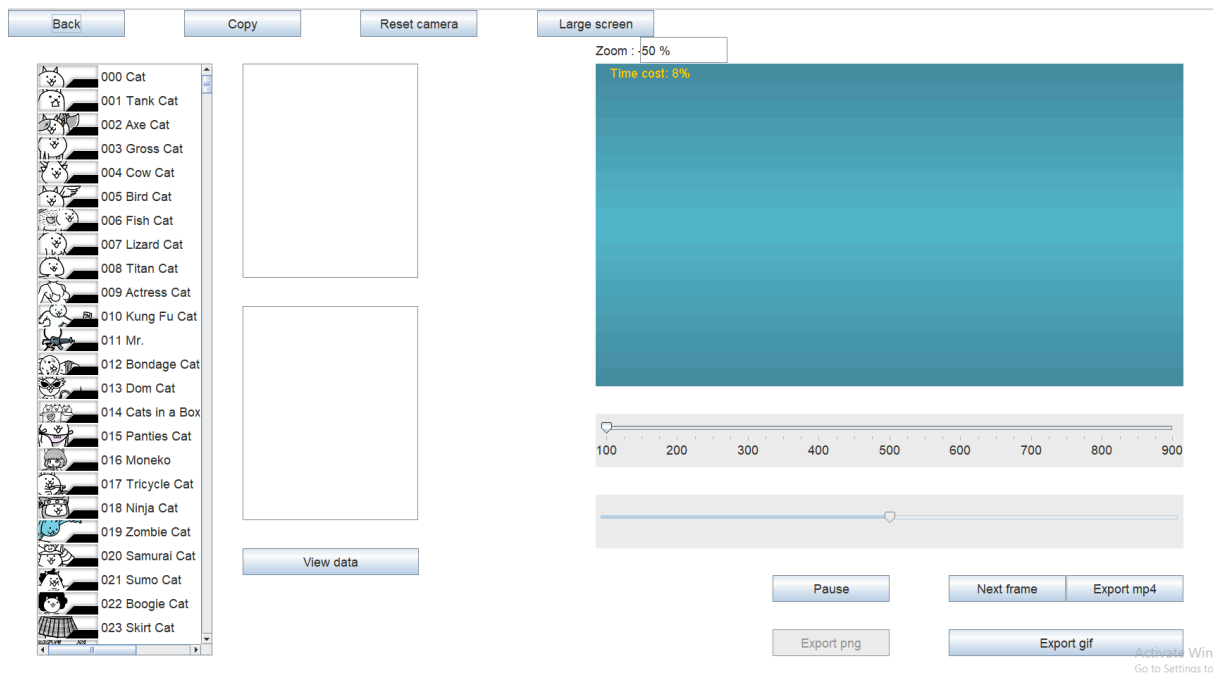
Part 3. Unit/Enemy animation

3.1 How to see animation

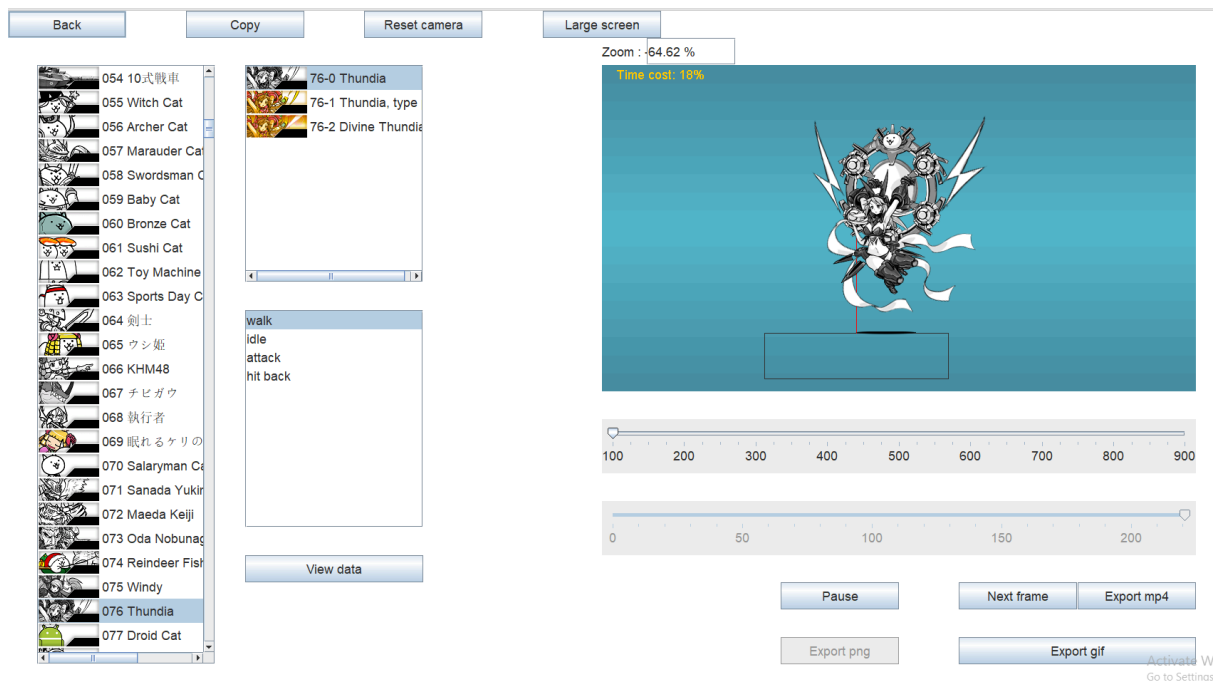
If you want to look at unit animations, click "unit anim".

If you want to look at enemy animations, click "enemy anim".

Then you will see a screen like this.



Click the unit/enemy you want to look at in the list on the left and BCU will show its animation. The units are listed by their internal IDs.



Unlike normal BC, it won't show different kinds of animations automatically. You need to select it below the form panel.

Scrolling up and down in the animation viewer to increase/reduce scale of unit/enemy and drag your mouse to move unit/enemy.

You can click the "pause" button to pause animation and "next frame" to see the animation of the next frame. Be aware that the "next frame" button will work only when animation is paused.

There are 2 control progress bars between buttons and the animation viewer.

The upper one can control the speed of animation. The lower one can only be controlled when animation is paused. You can see the specific appearance of units/enemies at a specific frame by controlling this bar.

You can also use your mouse and drag the top bar with the hundreds to increase or decrease the animation time.

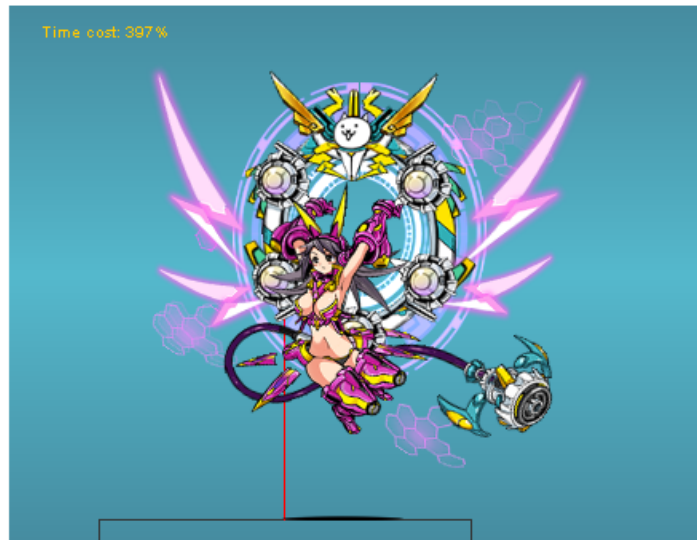
3.2 Exporting png and mp4

Now let's talk about exporting png and mp4, it is pretty simple.

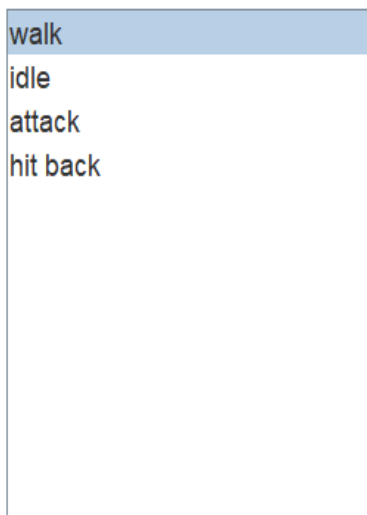
I'll talk about exporting to mp4 first.

For example, let's assume that you want to make a mp4 file of Thundia's attack animation.

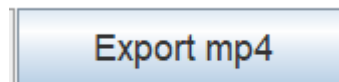
First, drag her to the proper position.



Then, go to the animation type, and choose attack.



After that, find the right starting time, and press the "export gif" button.



This button's name will be changed to "remain : xx". "xx" means the number of frames of the mp4 file. You can end recording by clicking this button again, and "remain" will start to reduce.

You have to wait for "remain" to become 0 or the gif file won't be generated properly. If "remain : xx" is changed to "export mp4" again, congratulations! You successfully get a nice mp4 of Thundia's walk anim, share it with your friends!

But where are these gif files saved you may ask? If you look at your BCU folder, there will be a folder named "img". Every picture and mp4 will be exported here.

Name	Date modified	Type	Size
assets	5/4/2022 10:44 PM	File folder	
backups	6/23/2022 4:07 PM	File folder	
BCU_lib	5/22/2022 1:45 PM	File folder	
exports	6/23/2022 3:21 AM	File folder	
img	7/13/2022 12:25 AM	File folder	
logs	7/13/2022 12:18 AM	File folder	
packs	7/12/2022 2:11 AM	File folder	
user	6/23/2022 4:07 PM	File folder	
workspace	3/24/2022 5:03 PM	File folder	
BCU-0-5-1-15	7/4/2022 7:09 AM	Executable Jar File	1,676 KB
BCU-Initializer	2/8/2022 4:34 PM	Executable Jar File	17,744 KB

Exporting png is a little bit different. You need to pause the animation first.

After that click "export png" and you will be able to find your nice pic in the "img" folder.

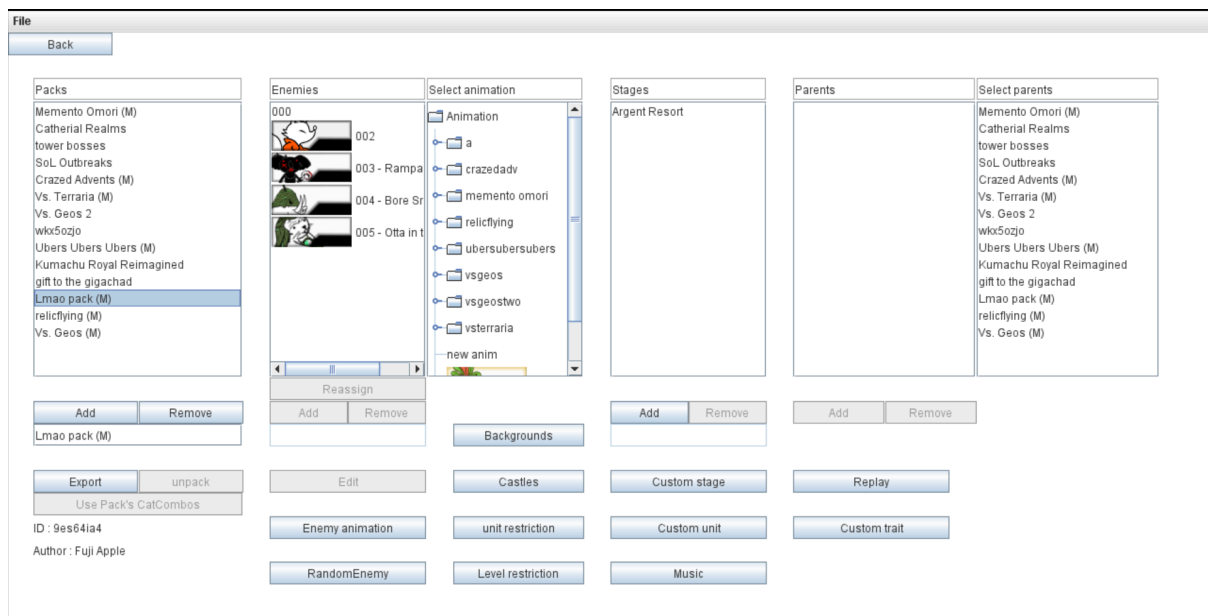
Let's start with the most important part in this tutorial.

Part 4. Custom Pack

Now we are going to make a nice custom pack. Are you ready for this?

First, if you want to play other people's packs, put the .bcuzip file in the "pack" folder in your BCU folder. If there's no such folder, create one. Note: Modern BCU no longer supports bcupack format.

4.1 - Interface



This is what you will see after clicking the button "custom pack" in the main menu. I'll explain what each part of this screen will do.

(1) is the list of your packs, choose one of them to check the detail of the pack

(2) is the list of custom enemies.

(3) shows all the custom animation you have. You can get them by making them or copy them from unit or enemy anim.

(4) is for custom stages, the detail of it will be taught below.

(5) shows the parent of your pack, you can use other people's packs into the custom stage by setting parent pack. Be aware that once you set parent packs, users will have to download parent packs to use your pack

4.2 - Making a Custom Level

Now let's make a level by ourselves

4.2.1 - Creating a pack

First, we need to create a pack.

Click "add" and a new pack file will be created. A new pack will be named as "custom pack". You can change the name of the pack by editing the texts in the textbox

Packs

Battle Cats: Pineapple Edition

Add

Remove

Export	unpack
Use Pack's CatCombos	

Add	Remove
tutorial	

Export	unpack
Use Pack's CatCombos	

I will use a pack named "tutorial" in this tutorial

4.2.2 - First Thing to Do

Go to (4)

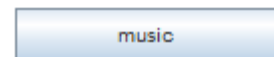
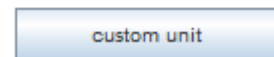
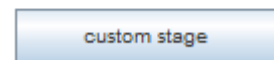
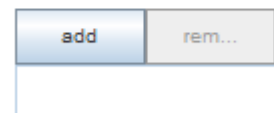
There are 2 things about custom stage
: Stage map, and stage

Stage maps contain stages. For example, in SoL, a stage map named "The Legend Begins" (Chapter 1) contains a stage named "Earthshaker".

To make custom stages, you have to create your own custom stage map first.

Click the "add" button and BCU will create a stage map named "new stage map". You can edit this name just like editing the name of the pack by editing texts in the textbox below the "add" button.

Then, click "custom stage".



Back

new stage map

Start

Add

Remove

Enemy data

000 - Doge
001 - Snache
002 - Those Guys
003 - Hippoe
004 - Pigge
005 - Jackie Peng
006 - Gory
007 - Baa Baa
008 - Sir Seal
009 - Le'boin
010 - Kang Roo
011 - One Horn
012 - Teacher Be
013 - Croco
014 - B.B.Bunny
015 - Squire Rel
016 - Assassin Be
017 - Shy Boy
018 - The Face
019
020
021 - Ms. Sign
022 - Mooth
023 - Dark Emper
024 - Teacher Bu
025 - 剣士
026 - ガウガウ
027 - ケリ姫

Base health

Length

Enemy Limit

Ch. cost

Background

Castle

Min respawn

Continuable

Music

Dojo Stage

Rarity

Normal

EX

Rare

Super Rare

Uber Rare

Legend Rare

First row only

Group

Level

Add line

Remove line

Limit

Replay

advance

Boss

Enemy

Hp / Atk

Amount

Base health

Start

Respawn

Layer

Group

Activate Windows

Go to Settings to activate Windows.

If you click it, this screen will appear

Select your stage map in the left panel, and click
"add" to add a custom stage.

Enemy data

000 - Doge
001 - Snache
002 - Those Guys
003 - Hippoe
004 - Pigge
005 - Jackie Peng
006 - Gory
007 - Baa Baa
008 - Sir Seal
009 - Le'boin
010 - Kang Roo
011 - One Horn
012 - Teacher Be
013 - Croco
014 - B.B.Bunny
015 - Squire Rel
016 - Assassin Be
017 - Shy Boy
018 - The Face
019
020
021 - Ms. Sign
022 - Mooth
023 - Dark Emper
024 - Teacher Bu
025 - 剣士
026 - ガウガウ
027 - ケリ姫

stage 0

1Star: 100%

2Star: /

3Star: /

4Star: /

Ch. cost

2

Base health

60000

Length

3000

Enemy Limit

8

Continuable

Background

null

Castle

null

Min respawn

1f

<0% health:

null

Music

null

<0% health:

null

Dojo Stage

Rarity

Normal

EX

Rare

Super Rare

Uber Rare

Legend Rare

Min cost: 0

Max cost: 0

Cat limit: 0

First row only

Group

null

Level

null

Add line

Remove line

Limit

Replay

advance

Boss

Enemy

Hp / Atk

Amount

Base health

Start

Respawn

Layer

Group

Activate Windows

Go to Settings to activate Windows.

Now there is a screen for making a level.

Choose an enemy from the panel on the left, and add it with "add line", and if you don't want that line anymore, choose that line and press "remove line".

But first, there are a lot of buttons and texts on top. The following explains most of them.

The textbox on left-top: The name of the stage

Text Boxes next to stage name: The strength of enemies for each stars

Base health: Health of enemy base

length: Distance between your base and enemy base

Max enemy: Max number of enemy

continuable: Continue Ability, if it looks like pressed, this level is continuable, but this won't affect anything to battle in BCU

background: Background of the stage, press the button and you can get a screen to choose bg, if you

memorized the number already then type it into the box next to it if you wish.

castle: enemy base of the stage, press the button and you can get a screen to choose castle, again if you memorized the number already then type it into the box next to it if you wish.

music: Music of the stage, press the button and you can get a screen to choose music (-1 means no music)

<(number)% health: Music will change when the enemy base has less than (number)% of health, and you can select the textbox in the right and press "music" to select music

rarity: Limitation of stage. Disable rarity to make users not able to spawn those types of units.

min cost: Minimum required cost that unit must have.

max cost: Maximum required cost that unit must have.

cat limit: Maximum number of cats in battle

loopstart: When the music in the stage is about to loop again, you can make it start with a specific time. Obviously when the stage starts the music will play straight away, but from the next loop, it will start playing at a specified time, this is read in milliseconds.

first line only: You can make users able to spawn units only in the first row of their lineup by enabling this button.

Group and Level are for restrictions for specific unit groups. You can adjust "group" to make users able to spawn only group units or make users not able to spawn group of those units. You can adjust "level" to make users able to spawn units which have a lower level than the restricted level.

4.2.3 - Enemies

Now, you know almost everything on the top, and let's talk about enemy things.

is boss	enemy	strength	number	base health	start	respawn	layer	group
		100%	1	100%	0	0	0~9	

Now let's talk about all the parameters! :D

is boss: Whether this enemy is boss or not, if it is, put any character in the textbox, if it spawns, it will use boss wave (pushing units back when it spawns) and it won't go behind the enemy base.

strength: Strength buff of enemy. (Default is 100%). You can set it as two values, (i.e. 150%,100%). The first value is health magnifier, while the second value is the attack damage modifier.

number: How many of this enemy will be summoned. You can make it blank or type 'infinite' to allow infinite enemy respawns, if it doesn't fit your enemy limit for the stage, then the enemy will be queued to summon when there's enough space for them to summon, let's say we got 4 capys but our enemy limit is 3, that 1 capy will spawn when you defeat a capy

base health: The enemy won't show up until the base health drops to the specific percentage. You can set it as two values with ~ to indicate the enemy will only spawn between those base health percentages. Set this to 0 and put the enemy as the first line to set it as animated base, which will set base hp to the hp value of said enemy.

start: The time (f, 30f is 1 sec) when the enemy firstly shows up. If you set this frame data as negative value, you can make it spawned synced with base health. If you set "start" as 30 normally, and "base health" as 90%, it will directly spawn when battle time passes 30f and base health becomes 90%. If you set "start" as -30, it will be spawned 30f later after base health becomes 90%. You can also use ~ to create random starting time

respawn: Enemies will respawn in every specific time decided by you. The ~ can be used to create random respawn times.

layer: There are walking paths in battle. 10 paths in total, you can decide the enemy's paths by editing this. This sets which enemy will go on top, 0 being the under everything else

4.3 - Custom unit/enemy

Making a custom unit is a fun thing to do, but this tutorial doesn't contain [animation tutorial](#), so I'll just teach you things about stats.

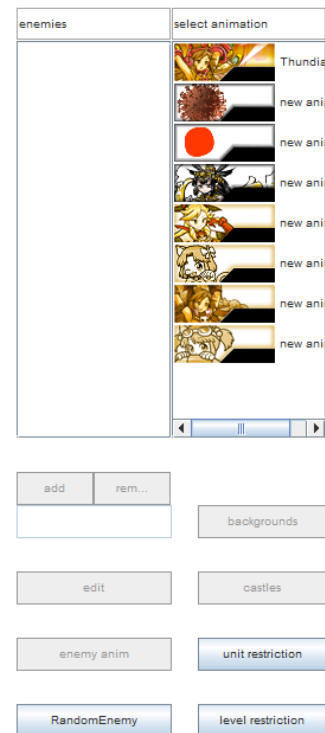
4.3.1 - First Thing to Do

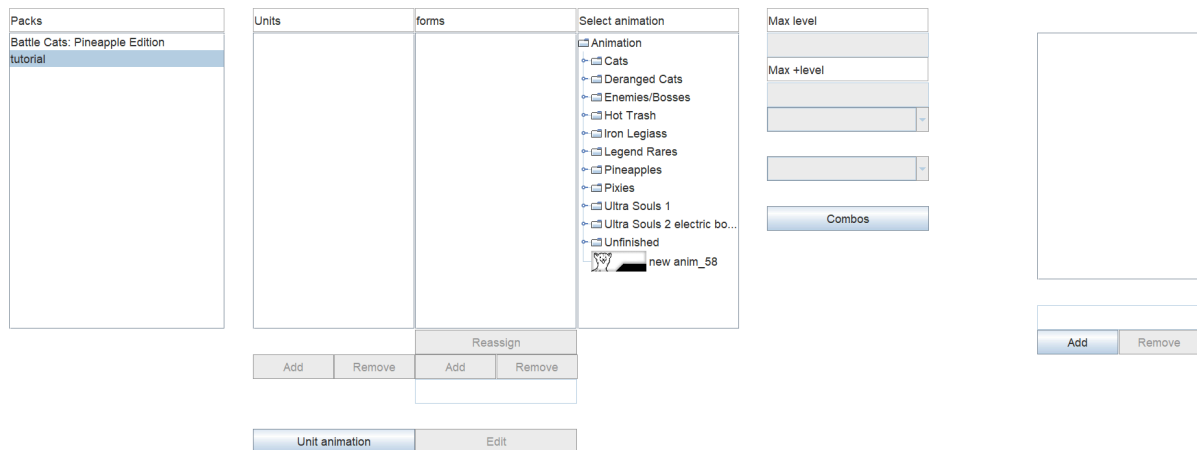
The picture on the right side are animations you made or you copied.

If you want a custom enemy, select the animation and click "add". Then you can change its name by the textbox below.

If you want a custom unit, you need to click "custom units".

After doing so, the screen below shows up.





Select an animation and click "add" below the units part, it will be the first form of that unit.

You can add other forms by just doing the same thing but click "add" below the forms part.

4.3.2 - Stats

Now let's talk about stats, after selecting the enemy or the unit, an "edit" button will light up. Click that button to edit that unit/enemy's stats/abilities

HP	10000	1		atk	0		Knockback			Summon			Berserker	
Hitback	1			Pre atk animation	1	chance	0		chance	0		health	0	
Speed	8			LD point near	0		Freeze			Moving Attack			Lethal Strike	
Willpower	1			LD point far	0	chance	0		chance	0		chance	0	
Width	320			type	1		Slow			Sniper Style Knock Back			Burrow	
Drop	0.0			dire	1	chance	0		chance	0		burrow count	0	
Range	0			count	-1		Critical Hit			Boss Shockwave			Revive	
TBA	0		Add Remove	ability	0	chance	0		chance	0		revive count	0	
To base	0		Copy Redirect	move	0		Wave			Stop Time			Immunity to Critical Hit	
type	1		common proc	Area attack		chance	0		chance	0		percent	0	
count	-1			DPS	0		Mini-Wave			Background Switch		Blocker	0	
				Post atk animation	20	chance	0		chance	0			Counter Attack	
				Atk frequency	21		Weaken		chance	0		chance	0	
				Gold Future_0		chance	0		chance	0			Invincibility	
				Enemy animati...			Barrier Breaker			Armor Break		chance	0	
				post-HB	x	chance	0		chance	0			Super Armor	
				post-death	x		Shield Piercer			Haste		chance	0	
				soul_0		chance	0		chance	0			Mystic Shield	
				Description Line 1										
				Description Line 2					Min position	-112.5				
				Description Line 3										
				Description Line 4										
star: 0			Import Units											

Enemy stat page(above), unit stat page(below)

Lv	40	CD	60	Layer	0~9									
HP	22000	1		atk	0		Knockback			Seal			Berserker	
Hitback	1			Pre atk animation	1	chance	0		chance	0		health	0	
Speed	8			LD point near	0		Freeze			Summon			Lethal Strike	
Willpower	1			LD point far	0	chance	0		chance	0		chance	0	
Width	320			type	1		Slow			Moving Attack			Burrow	
Price	75			dire	1	chance	0		chance	0		burrow count	0	
Range	0			count	-1		Critical Hit			Sniper Style Knock Back			Revive	
TBA	0		Add Remove	ability	0	chance	0		chance	0		revive count	0	
To base	0		Copy Redirect	move	0		Wave			Boss Shockwave			Immunity to Critical Hit	
type	1		common proc	Area attack		chance	0		chance	0		percent	0	
count	-1			DPS	0		Mini-Wave			Stop Time		Blocker	0	
				Post atk animation	20	chance	0		chance	0			Counter Attack	
				Atk frequency	21		Weaken		chance	0		chance	0	
				Gold Future_0		chance	0		chance	0			Invincibility	
				Unit animation			Barrier Breaker			Venom		chance	0	
				post-HB	x	chance	0		chance	0			Super Armor	
				post-death	x		Shield Piercer			Armor Break		chance	0	
				soul_0		chance	0		chance	0			Mystic Shield	
				Description Line 1										
				Description Line 2					Min position	220				
				Description Line 3										
				Description Line 4										
Import			Enemies Talents											

This is the screen you will see after clicking "edit." The following entries will explain everything. Entries marked with a (E) means that it is enemy exclusive, while entries marked with a (C) means that it is cat unit exclusive.

HP is the amount of hitpoint that a unit has. When this value reaches 0, the unit dies, and is removed from the battlefield. This can be decreased by taking damage from any source, or be increased if taking negative damage from any source.

Hitback, also known as knockback, is the amount of knockbacks the unit will take after reaching certain health percentages(calculated automatically). All units must have at least 1 hitback(for when they die). Any amount of hitback that is more than 1 will evenly split the max health of the unit to determine hitback percentages. For example, if a unit has 4 hitbacks, it will be knocked back at 75%, 50%, 25%, and 0% hp.

Speed is an arbitrary number that determines how fast the unit moves. A speed of 0 would keep the unit in place, despite showing the walking animation.

Willpower is the amount of space the unit will take in the cat/enemy limit. The default is 1, but can be set to any whole number greater than 0

Width is the size of the unit's hitbox. Almost all units have a hitbox size of 320.

(E)Drop is the amount of money dropped when an enemy is defeated.

(C)LV is the level of the cat unit's stats that you are creating. You may notice that health and damage are being rounded to a value, and that is because the stats must stay in the level curve. By setting the LV to a level, you are editing the stats of that cat unit at the specific level. BCU will scale the stats if you change the level again.

(C)Price is the cost to summon the cat unit in battle. This value is rounded.

TBA stands for "time between attacks", which means the period between the end of the previous attack and the start of the next one. For example, if TBA for a unit is set as 60f, the unit will wait 60f after the previous attack to start another attack. The unit can move forward during this.

Don't forget to set "to base". It is how far the unit will stand to the enemy base before it starts attacking the base.

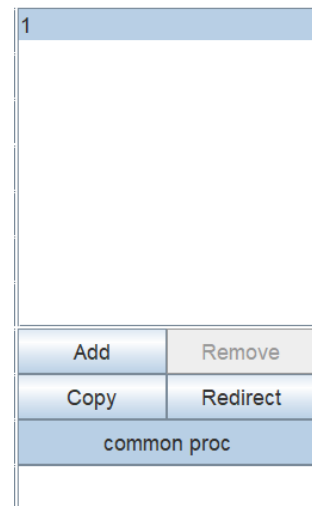
Type is the behavior in which the unit can target. By hovering over the field, the tooltip will state all of the + values. To include a type, simply add the value next to it to the total number.

Trait is the category in which the enemy is included in. It affects whether or not the unit targets(cat units) or is affected(enemies) by abilities with that trait. When you are selecting traits of enemies, you may be confused when trying to select multiple traits. Just hold ctrl while doing this.

You can copy some other unit's stats by clicking "import" and find the unit you want, click that and click "back"

This is the attack count area. The large white area shows how many attacks the unit will hit. Common proc here means that every attack will have the same abilities. Under that, the blank box is where you can name the attack. There are 2 special

names. By naming it "revenge", the attack will trigger every time the unit is knocked back from damage. By naming it "resurrection", it will be triggered every time the unit dies, even if not permanently.

The image shows a user interface for setting attack counts. At the top is a blue header bar with the number '1'. Below it is a large white rectangular area. Underneath the white area is a control panel with four buttons: 'Add' and 'Remove' in the top row, and 'Copy' and 'Redirect' in the bottom row. Below these buttons is a blue bar labeled 'common proc'. At the very bottom is a blank white box for naming the attack.

pre-atk(pre atk animation) is the number of frames of before attacking, unit/enemy won't do damage until this frame is passed. For example, if you set pre-atk as 30f, the unit will actually do the damage

on the 31st frame. **You must enter this manually to have it sync with your animation.**

post-atk is the number of frames after attacking, even though the unit/enemy already did damage to the target. If you set this post-atk to positive value, it won't move until this frame is passed. Let's imagine that pre-atk is 30f(1s), and post-atk is 30f(1s). It will take 30f to do damage to the target and another 30f to finish its attacking animation. For 60f(2 sec), the unit/enemy won't move. This total time (2 sec/60f) is called "atk-time". **Note: This is automatically calculated by BCU.**

To add a multi-hit attack, you have to click "add" next to TBA, and set the necessary parameters for that attack. If you want to set different abilities for each attack, disable "common proc".

I'll talk about some special data

Moving attack: This is an ability that creates an invisible hitbox that travels forward(speed). The size of this hitbox is set by width, and time is how many frames will the hitbox exist. Distance is how far away does the hitbox start from the unit. Interval is how fast the hitbox will deal damage/effects.

Theme attack: It can change background and music for a certain amount of time, and can be set to kill every other unit(both enemy and cat) when it triggers. Units with imu. theme will not be killed.

Sniper: It's like KB but a little bit different, just see when sniper shot an enemy

Revive: there is p0 and p1 in it, you may be confused about it but when you point your mouse on "type" you may find out some of the types requires range, p0 and p1 is the minimum and maximum range

Stop time: It stops time, which disables cash gain, prevents anything from moving(except imu. timestop)

Venom has 4 types of damage. This can be configured by changing the damage type value from 0-3; 0 being fixed damage(burn), 1 is % of total health(poison), 2 is % of current health(venom), and 3 is % of lost health(corrode). Time is how long the effect lasts, interval is how fast does each damage tick occur, and damage sets the damage.

In the

packs	enemies
000000 - default	
363008 - Covid19	
367738 - tutorial	 000 - dc
465680 - Absolute Bahamut	

In summon, you may be confused by the parameter "ID". If you want to summon units/enemies in original BC, use enemy data and check its ID. For custom enemies, check the number of pack and the unit/enemy. For example, the dot in the tutorial pack ID is 367736000. 000 is ID of the unit/enemy in the pack.

4.3.3 - Law of Unit Stats

When you are editing custom units, you may see it changes the numbers automatically, we will need the custom unit screen to explain that

Do you see {6, 8, 20} in the right?

That is level increment data.

ct animation	max level
 Thundia	50
 new ani	max +level
 new ani	0
 new ani	EX
 new ani	
 new ani	{6, 8, 20}

{6, 8, 20} means:

For 1~60 lv, HP and atk will increase 20% of stats of level 1 per level

For 61~80 lv, HP and atk will increase 10% of stats of level 1 per level

For 81~200 lv, HP and atk will increase 5% of stats of level 1 per level

After 200 lv, no stats will increase

I'll give an example with {6, 8, 20} data. This level increment mechanism is hard to understand.

Let's assume that a specific unit's stats of level 1 have 500 HP and 1000 atk. In level 2, unit's hp will be increased to $500 + 500 * 0.2 = 600$, and atk will be $1000 + 1000 * 0.2 = 1200$.

In level 3, unit's hp will be increased to $600 + 500 * 0.2 = 700$ and atk will be $1200 + 1000 * 0.2 = 1400$.

Keep going like this, and when unit's level becomes 60, unit's hp will be 6400 and atk will be 12800.

If unit's level becomes 61, then hp will be $6400 + 500 * 0.1 = 6450$, and atk will be $12800 + 1000 * 0.1 = 12900$.

And so on. Because of its complex calculation, there is a level increment calculator in this [link](#).

All unit's stats are decided by its level 1's stats, so even though you set atk and hp, values can be changed because of level increment.

4.4 - Custom Music, Backgrounds and Castles

This is almost the end of the tutorial, you are doing great, now let's talk about other custom things.

4.4.1 - Custom Castle

First, of course, select the pack.

Then, click "castles", after that, you'll see a screen.

click "add", select a file.

Yes, you finished it just that simple, now, check it out in a custom stage.

4.4.2 - Custom Music

Also, select a pack first and click "music".

At that screen, click "show", it will open a folder automatically.

Drag your music into that, make sure that your music is a .ogg file, and the name of the file is '000', '001' and so on.

You can use an online converter to change your music file into .ogg. Be aware that renaming extensions will never work. You have to use a proper converter to make it able to be played by BCU. It's not that hard to convert it since there are several online converters that are free and easy to use.

4.4.3 - Custom Background

Also, select a pack first and click "background".

press "add" on that screen and select a picture.

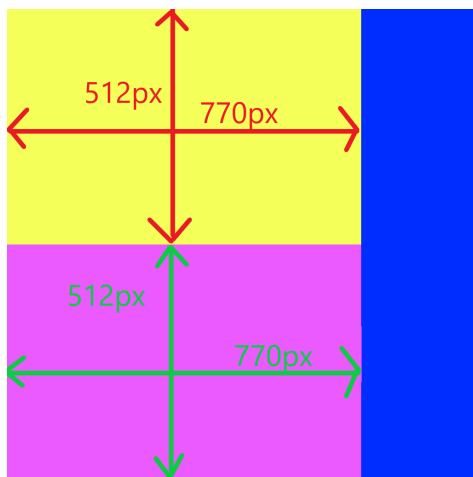
But be careful.

The picture should be a 1024*1024 file or 770*512 file, after that, result may be strange because it tries to crop the image into 770x512 subimages.

So the standard background size is 770x512.

If you have to put a large BG, you have to split it in 2 pieces and switch their position vertically, Both BG piece's size must be 770x512.

Let's use a pic as an example.



This is an example bg, There are numbers which show their length and width.

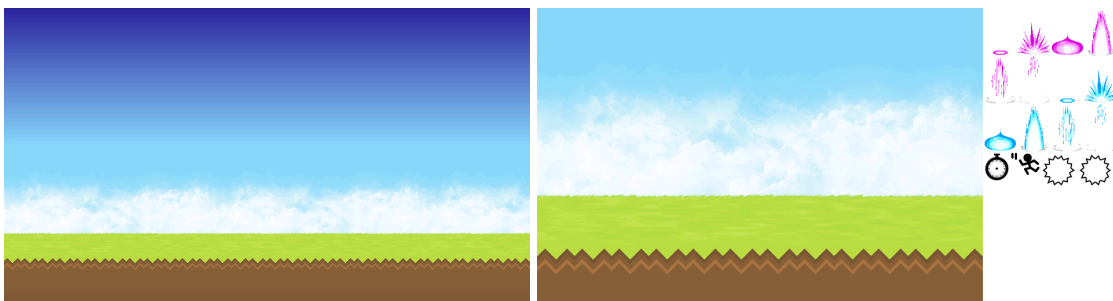
Then, after importing it, let's see the result.



This is what the real bg looks like.

As you see, the blue part will not be used and the yellow part and the pink part will be reversed.

the 4 textboxes above are about gradient color.

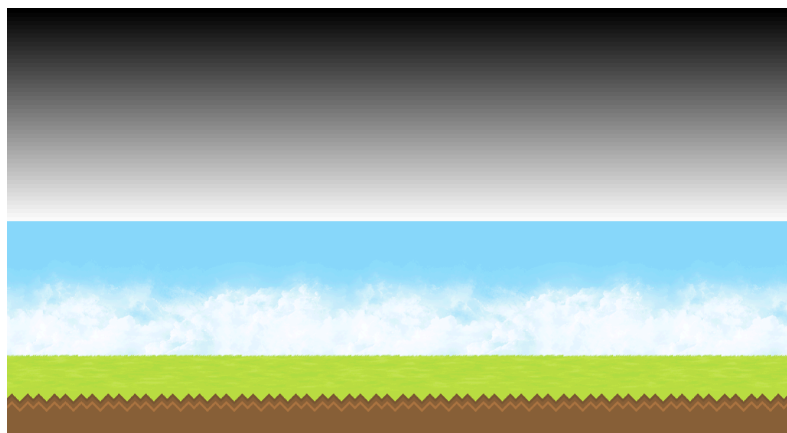


Left picture is the background in battle, and the right picture is its source image.

The smooth color in the sky in the left picture is called gradient color.

top	0,0,0	255,255,255
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Let's take an example. There are 3 numbers for each textbox. They are RGB value and order is (R, G, B). You can easily get RGB values with a color picker on an online site. (0,0,0) will be black and (255,255,255) will be white. Then below will be the result.
(Red, Green, Blue) if you didn't know



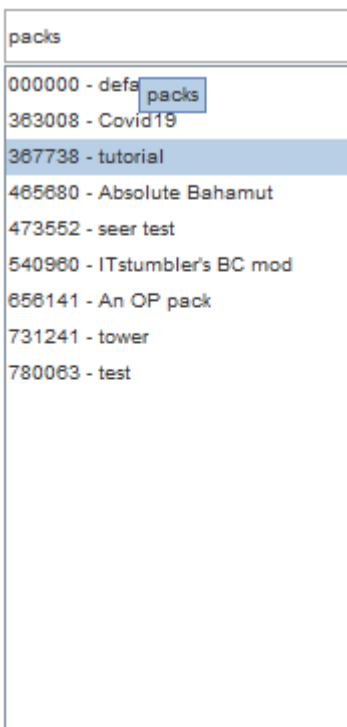
So first textbox means upper color of gradient, and second textbox means below color of gradient in sky.

Same applied to ground colors. Direction is a little bit different from sky colors, but the third textbox

handles the upper color of gradient, and the fourth textbox handles the color of gradient in ground. If you disable "top", the pink part will disappear. You can decide sky colors only when "top" is disabled.

Another detailed guide is in [this link](#) [Credit by Artherbit]

4.5 - Finishing Up



Now you know how to make a custom pack, click "export" and find your pack in the pack folder with the number.

In this case, "tutorial" pack's ID is 367738, so there will be "367738.pack.bcuzip" in the pack folder. This ID is unique, so you will have a

different ID from this pack, so you have to find it manually.

Have fun with it, hope this tutorial helped! :)